From Pre-Genesis to MSS-N

Jeremy H. Carroll

2026-10-09

Version 1.0.0

Abstract

An uninterpreted inscription supplies no admission policy, observation rule or mathematical realization on its own. This paper joins the recorded Pre-Genesis trials to the S(n) presentation account and the Minimal Symbol System atlas. It distinguishes supplied text, admitted objects, reports, grammar profiles and proposed capability targets. Exact report reconstruction, finite demand accounting, typed recovery and alphabet extension provide separate preservation criteria. At the chain-complex level, a retraction prevents an extension from killing an old nonzero homology class. Graph and word examples show why recovering syntax, distinguishing observations and repairing topology require different proofs. The companions construct integral component and triangle homology, canonical free-word reduction and an exact rational Bell-mixture obstruction. The three recorded model runs are historical examples with missing model metadata; they supply no success rates. The consolidation retains corrected source claims and withdrawn physical interpretations, and identifies the precise scope of its Lean, Python and Rust companions. As an entry companion to the Witnessed Difference Programme monograph, it prepares the report, recovery and repair contracts used by that work's later conditional finite-QM constructions.

1. Starting before <>

The first question is what has been supplied and what has been licensed to happen with it. The author, reader and computer already use a metalanguage. This paper makes that fact explicit; it does not describe an observer-free creation from nothing.

Condition	Host-level description	What has not followed
Absent input	No payload was supplied, for example None in a declared option type.	No object-level empty word or ground has been produced.
Empty word	The word epsilon in a declared free monoid has length zero.	It need not be accepted by a selected grammar start.
Literal <>	Two ASCII characters, possibly one opaque token under a stated lexer.	Neither an empty set nor a singleton carrier is forced.
Admitted ground	A named initial object under an explicit regime definition.	Its carrier, operations and permitted transitions still need definitions.

"Before" here marks the order of questions in the account. It is not a derivation of physical time. Even writing None presupposes a host capable of representing absence. Likewise, a grammar production cannot demonstrate that there is no grammar in the metalanguage in which that production was written.

Let a declared transition system have state set X and transition relation T. If T is empty, no state has a T-successor. This follows from the definition of an empty relation. It does not say that an external caller cannot provide more data, or that every later rule is prohibited. External supply and admission are events whose effects must be specified.

1.1 Reading toward the WDP monograph

This paper supplies the entrance to *The Witnessed Difference Programme — From Finite Distinction to Witnessed Measurement*. Pre-Genesis asks what was supplied, admitted and retained. S(n) separates presentation stages from mathematical degree. MSS-N supplies a vocabulary of proposed targets whose realizations need explicit contracts. The report and graph examples below make those contracts concrete before the monograph introduces its richer repair and observer models.

The contribution presented here is the organized passage from admission and observation questions to typed mathematical preservation and repair, with inspectable formal constructions and executable contracts. Standard algebraic results are identified at their points of use. The route is evaluated through its explicit maps, hypotheses and worked examples.

Read §§4–5 for sufficient reports, §§9–12 for typed recovery, and §13 for the preservation/repair distinction. Section 17.1 then gives the chapter handoff. The reading route and edition binding identify the companion draft. The later quantum carriers and readout rules are constructed in that monograph under its stated additional assumptions.

2. What the Pre-Genesis trials actually recorded

The Pre-Genesis strand contains the experiment design, Trial 1, Trial 2 and a full-history replay. Its trial record points to the original transcripts. The prompts withheld the intended comparison theories and asked the model to distinguish external presentation from internal expressibility. "No move is forced" was an admissible outcome. The next inscription was not prescribed as a successful rediscovery of an existing calculus.

The record supports the following limited observations:

Event	Recorded behavior	Evidential boundary
Bare marks and punctuation controls	A fenced prompt elicited familiar interpretations; raw controls elicited different replies.	The runs did not isolate every causal difference between conditions.
No declared continuation rule	All three runs said that a successor was not forced; some proposed a continuation under an added assumption.	A proposal and its adoption are different events.
`<a>` supplied after `<>`	Replies differed between presentation-relative and inscription-relative readings of identity.	The identity convention had not been fixed.

Event	Recorded behavior	Evidential boundary
A separation task about `<a>` and `<ab>`	A reply treated observational indistinguishability as identification; another used a stale account of the current state.	Lack of a distinguishing operation does not itself install an object-language equality rule.
Truncated audit versus replay	An auditor examining only the local response missed a historical-state mismatch.	Local consistency alone did not establish consistency with the adopted history.

Trial 1 also bundled steps and fenced controls that the design had intended to send separately and raw. These are recorded design deviations. The model identifier, settings and dates were not recorded. There were no host-checker verdicts on the replies and no live PFP evaluation. The earlier invented "50 trials" table is not part of the evidence used here. No rate, benchmark advantage or theory-rediscovery claim follows from this record.

The failure-mode chapter names semantic overcompletion, premature relation, false provenance, lost state, stale certification and bundled extension. These are retrospective descriptions. Naming a failure does not implement a detector for it.

3. Six operations that must remain distinct

The trials become easier to interpret when the following operations have separate signatures and records:

Operation	Meaning in a declared host
Mention	Refer to an inscription or possible object in a discussion.
Supply	Deliver a payload to an interface.
Designate	Identify which presentation is currently under consideration.
Admit	Add a datum, form, instance or rule according to an admission policy.
Retain	Preserve an admitted object or event for subsequent use.
Operate	Apply an admitted transformation with stated preconditions.

A response proposing `<a>` does not establish that `<a>` was adopted. A later task that mentions `<ab>` need not erase `<a>`. A checker cannot reconstruct these facts from punctuation alone; it needs event semantics and enough of the event record.

The trial method is consequently modest: keep discovery prompts free of expected solutions, preserve the exact controls, separate steps, record adoption explicitly, and bind an audit to its actual context. These are design lessons from the record, not experimentally established universal remedies.

4. Observation-relative difference

Let A be a declared carrier and O a family of functions on A. Write

$$x \sim_O y \quad \iff \quad o(x) = o(y) \text{ for every } o \in O.$$

Equality here is metatheoretic equality in each observation's codomain. If O is empty, every pair is equivalent. This does not imply that the inscription strings are equal or that the regime contains an equality operator.

Proposition 1 — separation cannot be recovered from a collapsed quotient. Suppose x and y have the same O-profile, while a task f requires f(x) different from f(y). No function on the current observational quotient can answer f on all inputs.

Proof. The quotient map q sends x and y to the same class. If f=g composed with q, then f(x)=g(q(x))=g(q(y))=f(y), a contradiction. Therefore a representation supporting this task must preserve more information than q preserves. This may be achieved by adding an observation that separates the pair. The proposition does not select that observation. □

For `<a>` and `<ab>`, a predicate for presence of b and a predicate for payload length greater than one both separate the pair, under a supplied string decoder. They disagree on `<aa>`. Full inscription identity also separates the pair and retains additional information. These are alternative admitted solutions, not capabilities obtained without a decoder or comparison operation.

5. A history projection can destroy the answer

Let H be a set of admissible histories, $\sigma : H \to S$ the state or protected answer required by a task, and $\pi : H \to V$ the report shown to an auditor.

Theorem 2 — history report criterion. An answer map $A : \pi(H) \to S$ satisfying $A(\pi(h)) = \sigma(h)$ for every $h \in H$ exists if and only if

$$\pi(h) = \pi(h') \quad \Longrightarrow \quad \sigma(h) = \sigma(h').$$

When it exists, it is unique on the report image. In particular, two histories with the same report and different required answers rule out an auditor that answers correctly on all admissible histories while seeing only that report.

Proof. A factorization gives $\sigma(h) = A(\pi(h)) = A(\pi(h')) = \sigma(h')$. Conversely, define $A(\pi(h)) = \sigma(h)$; the displayed condition makes this well-defined. Every member of $\pi(H)$ has this form, so the factor is unique. For a finite listed history population with decidable report equality, the construction is executable: find the first matching report and return that history's answer. Extending the map outside $\pi(H)$ requires a chosen answer there; no such extension is needed for reconstruction on admissible inputs. □

The carrier-general criterion and uniqueness on the report image are mechanized as PreMssn.history_report_criterion and history_factor_unique in the Lean companion. The existence direction uses Classical.choice; the finite listed-history reconstruction in the Python and Rust companions is constructive.

This is an information-loss theorem, not a theorem that every task requires every raw token of history. A sufficient state summary may preserve σ. Conversely, more context alone does not guarantee that an auditor will use it correctly. Raw history retention is a useful provenance policy; task-sufficient reconstruction is the mathematical requirement.

The local response in the trials could be consistent with its asserted premise and still be inappropriate for the adopted state. Checking internal steps and checking the binding of their premises to the actual history are separate tasks. The report criterion makes this distinction precise for any specified answer function on admissible histories.

6. The indices and layers

Symbol or object	Meaning	Separate quantity
S(n)	A presentation stage in the specified admission account.	The number of glyphs or a homotopy degree.
k	Alphabet size under a declared token policy.	Number of decoded objects or operations.
p	Grammar/profile and public start.	Semantic interpretation or evaluation verdict.
MSS-m	An externally specified atlas target.	A native Genesis stage or a completed lift.
H_j	Homology in degree j of a declared complex.	Stage n or active-demand count.
C(m), D	A declared coverage measure and length of a contiguous licensed chain.	One another; neither is inferred from the number of folders.

For example, S(2) in the teaching crosswalk records retained but observationally collapsed data. MSS-2 is a target about relations. Equating them by their numerals silently changes the object under discussion.

Four responsibilities remain distinct: native admissions; mathematical representations; downstream interpretations; and verification/evidence machinery. A Lean proof, a Python calculation and a SQL view may refer to related structures, but shared terminology does not prove that they evaluate the same property.

7. The finite-stage crosswalk

The Genesis account is the source for admission, observation, candidate costs and the sealed/post-seal distinctions. The following is a presentation crosswalk, not a claim that the displayed glyphs derive their own semantics or that every row is forced independently of policy.

Stage	Presentation question	Required qualification
S(0)	What is received as the opaque ground inscription?	<> has no inherited object-language meaning. Ground inertia is relative to the specified rules.
S(1)	Has a newly supplied mark been consumed or admitted?	Supply alone may be an internal no-op.
S(2)	What data have been retained, and which profiles remain equal?	Host-level multiplicity does not create an internal distinguishing context.
S(3)	How can a supplied pair-separation task be met?	The cost-two result uses a particular response grammar and cost; it is not a universal witness count.
S(4)	Have the relevant pairs been separated under the chosen closure policy?	Pair closure needs its finite-carrier and preservation assumptions.
S(5)	Where does the next requirement or fresh datum come from?	External continuation is a declared policy, not an endogenous generator proved by saturation.
S(6)	How are old presentations represented and recovered after an extension?	Typed maps and their preservation laws must be supplied and checked.

A branching set of tied responses is not automatically a linear tower. A chosen cost order, equivalence and continuation policy select or retain its branches. Subsequent atlas targets may ask for graphs, algebra, categories or geometry. Their presence in the atlas does not establish that the native process has reached them.

7.1 Where the HoTT interpretation entered

The Pre-Genesis trials did not derive homotopy type theory (HoTT) from the inscriptions. The S(n) account considered a connection with Postnikov towers, conditional on supplying a topological realization and its preservation maps. The MSS-N account explicitly introduced Genesis, HoTT and category theory as three mathematical lenses. Its types-as-spaces and identity-path vocabulary was an additional interpretation, not a conclusion of the trial record. The source reader identifies these historical passages.

The current Lean constructions concern ordinary equality, algebraic chain complexes and quotient groups. A chain homotopy is a relation between chain maps; it does not by itself construct a HoTT

identity type or a univalent universe. The corrected account therefore retains the mathematical connections while requiring a separate construction for any claimed derivation of HoTT.

8. Active demands, receipt sets and ordered histories

Let W_n be a finite register of active demands. A demand identifies its carrier, protected condition, witness data and activity predicate. It is not simply a label such as "curvature" or "missing structure." Let D_{n+1} be the demands discharged in a step and B_{n+1} the demands introduced. Assume

$$D_{n+1} \subseteq W_n, \qquad B_{n+1} \cap (W_n \setminus D_{n+1}) = \emptyset,$$

and define

$$W_{n+1} = (W_n \setminus D_{n+1}) \sqcup B_{n+1}.$$

Proposition 3 — finite demand accounting. With $s_n = |W_n|$, $d_{n+1} = |D_{n+1}|$ and $b_{n+1} = |B_{n+1}|$,

$$s_{n+1} = s_n - d_{n+1} + b_{n+1}.$$

Proof. Subset removal gives $|W_n \setminus D_{n+1}| = |W_n| - |D_{n+1}|$. Disjoint union then adds $|B_{n+1}|$. □

This identity does not prove which demands an engine actually carries. In particular, the previously displayed sequence (0,1,1,1,0,1,1) follows from one chosen assignment of singleton demands to the crosswalk. It is an illustrative ledger, not a universal native survivor sequence or a physical observable.

A cumulative receipt set $H_{n+1} = H_n \cup C_{n+1}$ has nondecreasing cardinality. An ordered event log additionally retains order and multiplicity. Neither is W_n. An empty active register need not erase receipts, imply thermodynamic equilibrium, or prove that every possible future question has been answered. Adding two map symbols does not imply two active demands or two newly admitted carrier elements.

9. Binary support and observation refinement

An inhabited irreflexive relation R(a,b) forces a≠b, by the irreflexivity argument below. Indeed, equality a=b would turn the witness into R(a,a), contradicting irreflexivity. This gives two endpoints for that edge, not two observers or two certificate occurrences. The pointed graph D2 with vertices 0 and 1 and the single edge 0→1 has exactly one endpoint-preserving map to a specified pair (a,b); that map preserves the edge. This elementary pointed property does not make D2 initial in an unspecified category of all possible systems.

Suppose q:A→O identifies a and b. If a supplied tag t:A→{0,1} has t(a)=0 and t(b)=1, define q'(x)=(q(x),t(x)). Then q'(a)≠q'(b), while projection to the first component recovers q. Both claims follow directly from equality of ordered pairs. The tag and its construction are supplied data. A method for finding tags is a separate algorithmic obligation.

For several required pairs, form a conflict graph on A. A tag assignment separating every edge is a proper coloring. Thus the number of tag values needed for many pairwise constraints need not

remain two: a three-clique needs three colors. The constraints remain binary without implying a globally binary solution space.

10. Typed recovery on the old image

Let u:A→B be a proposed lift and let

$$\hat{r} : B \to A + \{\text{NON_DESCENT}\}$$

be a typed recovery map. The old-image law is

$$\hat{r}(u(a)) = \text{inl}(a) \quad (a \in A).$$

Proposition 4 — injectivity. This law implies that u is injective. If u(a)=u(a'), applying the recovery map gives inl(a)=inl(a'), hence a=a'. □

The law alone says nothing about off-image inputs. A checker intended to reject all of them needs the additional specification

$$b \notin \text{im}(u) \implies \hat{r}(b) = \text{inr}(\text{NON_DESCENT}).$$

It also needs a way to decide the image condition within its declared domain and budget. That is immediate for some finite presentations and not a general computability theorem.

Two further qualifications prevent stronger promises from being inferred:

- **Observation preservation needs a commuting condition.** For old observation q:A→O and new observation q':B→O, require q' composed with u=q, or provide a declared transport to the old report space. Recovering a term does not by itself prove that a new evaluator uses the old observations.
- **Repair needs its own proof.** An identity embedding has a perfect old-image law and can leave every original obstruction unchanged.

The pair is not generally an involution. Moreover, u composed with r-hat is not even a map B→B until handling of the refusal summand has been defined. No global preservation of all certificates follows from a successful term round trip alone.

The current SnCarrier model restricts the section to BasePresentation: ground, supplied mark, retained pair, directed cut and simplex cut. Nested lifts, retractions and refusal tokens are outside this source type. The theorem SnTheorems.old_image_base establishes the recovery equation for every base constructor and every stage. `Carrier n` is `Fin (carrierCard n)`, and ground_carrier_empty establishes that the index-zero carrier is empty. These definitions correct the older unrestricted fallback and cardinality-table mismatch described by the predecessor draft. They establish the declared local model, rather than a correspondence theorem for every parser or source regime.

11. Formal languages and conditional expressive capacity

A symbol system must declare terminal spellings, tokenization, grammar, public start, layout policy, identity policy, annotation decoding and allowed manipulations. These eight choices determine what an inscription does.

For the familiar alphabet Γ_{38}, the glyph inventory comprises 26 lowercase Latin letters, ten decimal digits and <, >. Some MSS grammar profiles add punctuation or Unicode; they are not confined to these 38 characters.

Keep three relations separate:

$$E \subseteq V \times V, \qquad \rightarrow_{\text{rewrite}} \subseteq \Sigma^* \times \Sigma^*, \qquad q : V \to O.$$

They represent an encoded relation, rewrites between inscriptions and an observation map respectively. A change in any one need not change the others.

With repetition admitted, the one-letter language a^+ supports positive unary counting under the length decoder. Including epsilon gives a^* and a representation of zero. With two letters, concatenation is noncommutative: ab and ba are different words. A declared binary decoder can use any two distinct printed letters. A relation may be encoded by positions or a table without an arrow glyph. These are conditional constructions, not semantic resources supplied by alphabet size alone.

11.1 Free monoids and numeral decoders

Theorem 5. Free monoids on finite alphabets of different sizes are not isomorphic.

Proof. In a free monoid, the nonidentity elements that cannot be factored as a product of two nonidentity elements are exactly the one-letter words. A monoid isomorphism preserves identity and this indecomposability property in both directions. It therefore induces a bijection of the generating alphabets. Their sizes must agree. □

Thus binary and decimal string monoids are not isomorphic. Decoders can map their strings onto the same natural numbers without making the string monoids identical. A fixed-width binary encoding of decimal digits is injective under concatenation, but not onto the binary free monoid.

Likewise, discrete categories with two and 26 objects are not equivalent: all isomorphism classes are singletons, and an equivalence would have to match those classes bijectively.

11.2 Three recovery constructions with different types

For alphabets A contained in B, language inclusion and restriction are

$$\ell : \mathcal{P}(A^*) \to \mathcal{P}(B^*), \quad \ell(L) = L, \qquad r(M) = M \cap A^*.$$

Then $r(\ell(L)) = L$. This is a retraction of sets of words.

For individual words, exact recovery of old words can instead return the same word if it is in A*, and a typed refusal otherwise. A total erasure homomorphism h:B→*A* is another choice: map each new letter to epsilon and concatenate. It recovers every old word but loses new letters and their positions. These are three distinct contracts, not one overloaded "down-map."

No abelian category of ordinary formal languages has been specified here. An Ext calculation cannot be inferred from language restriction. Even in an abelian category, a split extension is one zero extension class; it does not show that an entire Ext group vanishes.

12. Manipulations and actual information loss

A renaming f:A→B has a two-sided inverse on the entire codomain B exactly when it is bijective. Recovery of source letters from the image f(A) only requires injectivity. Mapping both c and d to d collapses ac and ad. This is independent of whether the resulting word is grammatically well formed.

For the code a↦0 and b↦00, the different source words aa and b both encode as 00; ab and ba both encode as 000. A decoder cannot recover the source universally without additional retained information or a restricted source domain.

For declared directed-edge syntax:

Inscription	Represented edge
`a>b`	a→b
`b>a`	b→a
`a<b`	b→a
`b<a`	a→b

Reversing the vertex sequence and every arrow in `a>b>c` gives `c<b<a` and preserves its directed edge multiset. It changes the traversal order. A checker must name whether it protects the edge multiset, the ordered trace, annotations, or exact source bytes. AST recovery and byte recovery are different questions.

13. Finite graphs and the homological obstruction

For a finite directed graph, regard the underlying graph as a one-dimensional cell complex and define

$$C_1 = \mathbb{Z}^E \xrightarrow{\partial_1} C_0 = \mathbb{Z}^V, \qquad \partial_1(e) = t(e) - s(e).$$

If V is nonempty and the underlying undirected graph has c components, then $H_0 \cong \mathbb{Z}^c$ and $\widetilde{H}_0 \cong \mathbb{Z}^{c-1}$. The maximal component count |V| occurs when no edge joins distinct vertices; self-loops do not affect it. Adding an edge reduces the component count only when it joins previously distinct components. It does not separate a and b under an unchanged observation q with q(a)=q(b).

These standard graph and homology facts use the conventions in Hatcher's *Algebraic Topology*, §1.A, Proposition 2.7 and §2.A.

Formal negative coefficients do not authorize reverse physical traversal. In a CW graph, a self-loop has zero boundary and can generate nonzero H_1. It is not a degenerate simplex solely because its endpoints coincide; the model matters.

13.1 The chain-retraction obstruction

Theorem 6. Let $i : C \to D$ and $r : D \to C$ be chain maps with $r \circ i = \mathrm{id}_C$. Then $H_j(i)$ is injective for every j. In particular, i cannot kill a nonzero homology class of C.

Proof. If z is a cycle in C and i(z)=partial_D b, then

$$z = r(i(z)) = r(\partial_D b) = \partial_C r(b).$$

So z was already a boundary in C. Therefore the kernel of $H_j(i)$ is zero. □

This is the central preservation limit inherited from the MSS-N monograph. It does not prohibit every useful extension; it identifies incompatible promises. An extension may repair a cycle while preserving inscriptions, labels or selected questions, but it cannot simultaneously claim this chain-map retraction fixing all of C.

The obstruction persists when the retraction equation holds only up to chain homotopy. The homotopy companion establishes that homotopic chain maps induce the same map on the constructed homology quotient. A homotopy left inverse therefore still induces a left inverse and forces injectivity. In particular, the filled triangle below has no such homotopy retraction onto its unfilled complex.

13.2 Two isolated points

Let $C_0 = \mathbb{Z}a \oplus \mathbb{Z}b$ and $C_1 = 0$. Add an edge e with $\partial_D e = b - a$, obtaining D. A retraction fixing the original vertices would have $r_0 = \mathrm{id}$ and $r_1(e) = 0$. The chain-map equation would require $b - a = \partial_C(0) = 0$, contradicting the linear independence of a and b in the free abelian group. The new edge kills the old reduced zero-dimensional class, so the theorem applies.

13.3 A triangle and an explicitly attached face

Under label identity, `a>b>c>a` represents edges e_{ab}, e_{bc}, e_{ca}. Their sum z has boundary $(b - a) + (c - b) + (a - c) = 0$. With no two-cells, H_1 is generated by $[z]$. Adjoining a face f with $\partial_2 f = z$ makes z a boundary and satisfies $\partial_1 \partial_2 = 0$. The inscription gave the one-skeleton; the face is an additional declared object.

Killing this class prevents a chain-map retraction fixing the original complex. It does not prove that an unrelated observational collision has been resolved. Nor does one vanishing class show that every homology group vanishes.

Chains and cochains must remain typed separately: $C^j = \mathrm{Hom}(C_j, \mathbb{Z})$ and $(d^j\varphi)(c) = \varphi(\partial_{j+1}c)$. Filling a chain is not the same equation as finding a cochain primitive. Homological vanishing also differs from null-homotopy: the commutator loop in the wedge of two circles has zero abelianized homology class but is nontrivial in its free fundamental group.

13.4 Constructed homology groups and integral classes

The homology companion constructs the cycle subgroup, the equivalence relation modulo boundaries and its quotient abelian group. It constructs the homomorphism induced by a chain map and establishes the induced retraction equation. The theorem induced_homology_injective realizes Theorem 6 on these quotient groups. Group laws, well-definedness and injectivity are proved from

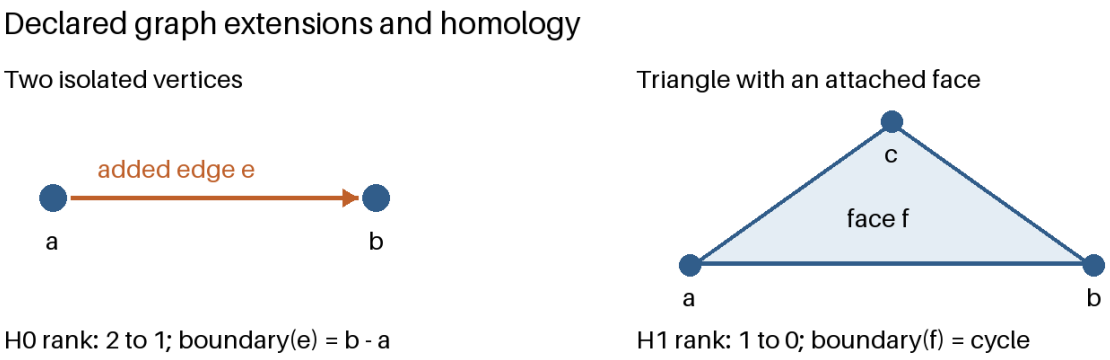


Figure 1: Two explicitly declared graph extensions. Killing the old class prevents a chain-map retraction fixing the old complex.

the supplied additive-group and chain-map laws; they are not assumed properties of a named homology object.

An integral test complex has three consecutive chain groups equal to $\mathbb{Z}$, with $\partial_2(k) = mk$ and $\partial_1 = 0$. Its middle classes satisfy $[n] = [p]$ exactly when $n = p + mk$ for some integer k. The companion establishes that Euclidean remainders classify these classes for every integer m, including m=0. At m=2, the class of 1 is nonzero and its double is zero. This is a constructed torsion example, without assigning it to an undeclared surface. The shared runtime encoding retains n when m=0 and otherwise returns the nonnegative remainder modulo $|m|$.

13.5 The component group is constructed

The graph companion constructs integral zero-chains as the free abelian group on vertices and one-chains as the free abelian group on edges. Its boundary sends an edge to its target minus its source. Connected components are the quotient of vertices by the equivalence relation generated by edges, allowing reversal and concatenation of paths.

The maps in h0ComponentIso give mutually inverse homomorphisms between the boundary quotient and the free abelian group on these components. The construction also identifies that boundary quotient with the kernel/image homology group of the graph's degree-zero chain fragment. Thus the unreduced component statement in §13 has a carrier-general companion, including empty graphs, loops and parallel edges. The free abelian group's extension and uniqueness properties are established from its term quotient; freeness is not a supplied hypothesis.

The theorem vertex_class_eq_iff_connected states that two vertex generators give the same zero-dimensional class exactly when they are connected. A loop's boundary is zero. In the finite replay, a component is named by its least vertex and a zero-chain is sent to its integer coefficient sum in each component. These coordinates preserve component information and vanish on boundaries; they do not alter the observation map q.

13.6 Finite coordinates and constructive edge retractions

The finite-coordinate construction identifies the free abelian group on a carrier enumerated by `Fin c` with integer c-tuples. It transports this isomorphism to the actual graph component group. For c=n+1, the augmentation kernel is isomorphic to integer n-tuples: retain the last n coordinates and recover the first as minus their sum. This establishes the displayed reduced-rank formula from explicit inverse homomorphisms.

The edge-extension construction establishes that adding an edge reduces the finite component count exactly when its endpoints were previously disconnected. It also establishes the maximal-count criterion and that a loop gives a nonzero class in the graph complex with no two-cells.

More precisely, an added edge has a chain retraction fixing every old zero-chain if and only if its endpoints were already connected. In the connected case, the construction sends the new edge to an integral old path chain with the same boundary and fixes the old edges. In the disconnected case, its induced homology map identifies two old distinct vertex classes, contradicting Theorem 6.

The component algorithm merges label fibers along the supplied edge list. Its labels agree exactly on connected vertices and, equivalently, on equal vertex classes in H0. Choosing the first matching vertex in a complete enumeration gives actual component representatives and a basis for H0. With vertices enumerated in increasing order, the construction establishes that the selected representative is the least vertex in its component.

13.7 Triangle classification and the nonabelian distinction

The triangle construction establishes that every integral cycle has a unique form (k,k,k). Its actual unfilled H1 quotient is isomorphic to the integers. Every cycle becomes a boundary after the displayed face is attached, and the filled H1 quotient is trivial. The attachment kills the old generator and has no chain retraction fixing the old middle group. Noncycles are refused by the runtime class encoder.

More generally, prescribe the face boundary as m times the triangle cycle for any integer m. The degree construction gives actual inverse homology maps to the integral quotient complex, so its H1 classes are classified by integers modulo m. At m=0 the full integer coordinate remains; at m=1 the group is trivial; at m=2 the cycle has a nonzero order-two class. This is an algebraic chain-complex construction, without an inferred surface realization.

The dual construction forms cochains as additive homomorphisms and coboundaries by precomposition. The chain square-zero law implies the cochain square-zero law. An edge cochain with coefficients (p,q,r) evaluates on the attached triangle face as p+q+r. Pullback along a chain map reverses its direction. A chain retraction gives a split surjection on cohomology in the inclusion direction, as the dual construction establishes for arbitrary abelian coefficient groups.

The free-group construction is a term quotient by the group laws with a proved unique homomorphism extension property. Its commutator on two generators is nontrivial, as a finite permutation interpretation detects. Abelianization has the unique factorization property for maps into abelian groups. The reduction theorem strengthens this witness: two signed words represent the same group element exactly when adjacent inverse cancellation gives the same reduced word. The reduction algorithm terminates structurally and is idempotent.

The two-loop chain complex has H1 isomorphic to integer pairs. The commutator is nontrivial in the constructed free group and has zero class in this actual homology group. These algebraic

and cellular constructions do not yet identify that free group with the quotient of continuous based loops by homotopy in a topological two-circle wedge. That comparison remains the cited background statement and an open formal obligation.

14. Profiles, parsers and evidence scope

The MSS grammar defines the 22 explicit starts inventoried in Appendix A. It supplies formation rules, not physical models or admitted semantic operations.

Selected profile	`a>b>c>a`	`a0>b2>c1>a0`	Interpretation if a decoder is supplied
raw_glyphs	Accepted	Accepted	Character sequence only.
label_path	Accepted	Accepted	Whole labels such as a0 are identifiers.
suffix_path	Rejected	Accepted	A letter plus a required digit suffix.
glyph_path	Accepted	Rejected	Single glyphs separated by direction marks.

An outer enclosure is not a universal requirement. Profile choice is part of the input contract; a parser must not guess a favorable interpretation after seeing an otherwise invalid candidate. Byte sequences, Unicode scalar values, grapheme clusters and grammar tokens also need separate counting conventions.

The historical inventory describes 22 starts alongside six retained runtime profiles. That inventory is not evidence of 28 semantic evaluators or a single shared resource limit. Recognition, AST construction, decoding, evaluation and exact serialization are distinct checks. A scalar syntax that accepts `a:1/0` still needs a decoder to reject zero denominators unless the grammar itself has been changed to exclude them.

The historical source manuscripts preserve the original parser descriptions and historical codec example. They must be bound to their profile and implementation revision before being used as current protocol requirements.

15. What a justified lift must demonstrate

For a claim that an extension is necessary or minimal:

1. State the failed predicate and protected observations or laws.
2. Specify admissible repairs within the current carrier and operations.
3. Establish that those repairs cannot meet the predicate, or report incomplete search if the budget prevents that conclusion.
4. Specify the extension class, equivalence and cost order.
5. Retain all tied minima unless an explicit selection policy is applied.
6. Check that each accepted extension meets the predicate and preserves the particular contract being claimed.

A finite candidate census is exhaustive only for its declared domain. A budget stop is not proof of obstruction. A declared successor, a proposed lift and a licensed transition therefore have separate evidence states. Exact names of runtime terminal values belong to the engineering protocol's schema; combining papers does not create a new enum or claim every runner already conforms.

Similarly, proposer, evaluator and authorizer are roles with different inputs and responsibilities. Three agents are neither necessary nor sufficient. A theorem that an already halted state is unchanged by replay establishes that equation; it does not prove the authorization of an active transition.

16. Boundaries retained from the corrected walkthrough

The following withdrawn claims must not re-enter through consolidation:

Earlier inference	Correct scope
Two-sided inverse carrier maps prove univalence.	They establish the stated equivalence. A univalent universe requires a different construction and proof.
S(n) is automatically a Postnikov tower.	Construction stage and truncation degree differ; maps and preservation properties of a fixed-space Postnikov tower are extra data.
A selected CHSH expression detects all contextuality.	All required orientations and the model assumptions must be checked. A Bell score alone is not a constructed Cech cohomology class.
$\texttt{F}^2\texttt{=I}$ establishes a pentagon identity.	It is an involution check; full associator coherence is a different statement.
A SQL comparison and a Lean compilation prove bisimulation.	A relation connecting their states and transitions still needs a proof.
Small matrix/graph fixtures certify a physical theory.	They establish only the checked finite calculation, within their arithmetic model and input assumptions.

For the correlator tuple $(1, 1, -1, 1)$, the selected expression $E_{00} + E_{01} + E_{10} - E_{11}$ is zero, while $E_{00} + E_{01} - E_{10} + E_{11}$ is four. Uniform-marginal pairwise distributions with those correlations exist. No global deterministic assignment can satisfy all four signs: the product of the four required pair products is -1, but every assignment gives $+1$. This is the concrete missed orientation from the corrected walkthrough.

The pair-distribution construction supplies probabilities 0 or $1/2$, establishes normalization and both uniform marginals, and computes all four correlations exactly. The mixture theorem establishes that every normalized nonnegative rational distribution over the sixteen joint deterministic output assignments has oriented CHSH score at most two. The proof checks the deterministic bound and sums it with nonnegative weights. Hence no such joint distribution has correlations (1,1,−1,1), whose oriented score is four. This is a statement about the declared finite probability model; it supplies no physical realization.

The correction record records removed fabricated implementations and repaired finite examples. Valid trace, determinant, commutator, boundary and finite graph calculations remain useful. Their

utility does not reinstate removed CDT, Super-WDW, sign-problem, chiral-superconductor or universal physical-verification claims. The verification scope of this paper is stated below; it does not restore those withdrawn claims.

17. A common preservation question

The report criterion in §5 connects the examples without identifying their types. For a report q and protected answer f, reconstruction on the report image is possible exactly when f is constant on each fiber of q. A grammar decoder, history projection or observation map can therefore be checked against a named question. A typed recovery map asks for an inverse on a declared source image; a chain-map retraction asks for stronger algebraic compatibility. Each promise must be stated before an extension is evaluated.

An atlas heading is a proposed capability. A candidate lift is submitted data. A verified construction meets a particular mathematical contract, and permission to use or publish it is a separate decision. These distinctions do not establish a native derivation of every MSS capability from the initial inscription.

17.1 Handoff to the Witnessed Difference Programme

The monograph continues the preservation question using declared repair maps, observer interfaces and progressively richer carriers. The following route identifies the assumptions passed forward rather than identifying objects because their names agree. It targets the companion edition identified in the reading route.

This paper prepares	Monograph destination	Data carried into the next model
Supply, admission, stage and retained history (§§1–3, 6–10)	Chapters 1–3; Appendix L	Actual admitted objects, permitted changes and source provenance
Report-image factorization (§§4–5)	Mathematical entrance; Appendices P.1 and K.1	The report, protected question and decoder or collision witness
Chains, dual cochains and retractions (§13)	Chapter 4; Appendices P.2–P.4	The specified complex, repair image and commuting transport maps
Words, reversal and signed generators (§§11–13)	§§17.7–17.8	Typed histories, protected contexts, admitted signed completion and reversal
Pairwise probabilities and the joint-model obstruction (§16)	Chapters 18–19; Appendix J.4	Explicit comparison models and the chosen composition/readout assumptions

The next steps toward finite QM are the monograph's conditional constructions: §17.9 supplies a checked complex structure and Hermitian form; §17.10 supplies state/effect certificates and an admitted trace readout; §17.11 supplies process families and composition. This paper prepares their

information and preservation questions. Those additional structures are not selected by the initial glyphs, alphabet size or a successful recovery round trip.

17.2 One concrete passage from graph syntax to repair

Supply the oriented triangle and an integral edge cochain h=(p,q,r). Its circulation is L(h)=p+q+r. Declare vertex-potential repairs

$$d_0(a, b, c) = (b - a, c - b, a - c).$$

Then L(d_0(a,b,c))=0 and L(h-d_0(a,b,c))=L(h). Conversely, L(h)=0 permits the potential (0,p,p+q), whose repair is exactly h. Thus circulation is the complete obstruction to repair by these declared integral potentials. The cochain (1,1,1) has circulation three and cannot be repaired to zero this way. The cochain companion establishes both the invariant and the exact repair criterion as triangle_repair_invariant and triangle_repair_exact_iff.

This hands a specified repair problem to Chapter 4 of WDP. It differs from attaching a face to kill a chain homology class: the carrier and repair map here are declared cochain data. A norm and optimization problem are further inputs; the monograph develops their exact certificates. No physical interpretation is needed for this handoff.

Mechanism inventory

Mechanism	Mathematical evidence	Executable scope
Pre-Genesis trials	Three source-recorded runs; explicit separation and history questions	Exact primary records and grammar controls; no benchmark or learning claim
Report reconstruction	Constancy on fibers, factorization on the image and uniqueness	Carrier-general proof obligations and finite report counterexamples
Presentation stages and demand registers	Supplied stage crosswalk and finite removal/disjoint-birth identity	Ledger examples are fixtures rather than a forced native sequence
Typed old-image recovery	Injectivity from the old-image law; base-presentation recovery	Current SnCarrier definitions; no blanket parser/source refinement claim
Words and language extension	Free-monoid generator distinction, inclusion/restriction and encoding collisions	Declared finite alphabets, token policies and exact word examples
Chain retraction and graph examples	Constructed homology groups, free component group and induced-map injectivity	Declared chain groups; exact graph/triangle, component and integral quotient examples
MSS grammar profiles	Twenty-two named starts with separate decoders and interpretations	Recognition, AST construction and semantic evaluation remain distinct
Lift selection and authorization	Explicit repair domain, cost, ties and protected conditions	Budget exhaustion is not proof; an execution result grants no authority

18. Sources and current verification status

The source reader and source manifest bind 34 imported files to exact Git blobs at revision d1 d2f034198a72b730320a69691e63351557c019. The internal intake also preserves the predecessor manuscripts and historical review records. The consolidated paper replaces the former education paper packages; the Pre-Genesis experiment records remain in their source location.

The standalone package builds with Lean 4.33.0 and no external Lean dependencies. Its complete compiled-declaration census includes generated declarations and records individual dependencies on propext, Quot.sound and Classical.choice. The report criterion, demand accounting, typed injectivity and finite-alphabet non-isomorphism statements are in lean4/Paper.lean. At the chain level, the companion formalizes cycle preservation, boundary reflection and reflection of the relation defined by a boundary difference. lean4/Homology.lean additionally constructs quotient homology groups and establishes injectivity of the induced map under a chain retraction. It also establishes the integral class-encoding criterion and the nonzero order-two example. The graph homology module constructs the unreduced graph component group and identifies vertex-class equality with connectivity. The finite-coordinate, reduced-H0, triangle-H1, constructive edge-retraction and free-group results are constructed in the companions described in §§13.6–13.7. Continuous topological

fundamental groups and their comparison with the displayed free group remain outside the current formal construction.

The observation companion constructs the profile setoid, its quotient, the task-descent criterion and the unique pointed D2 map. The word-recovery companion constructs fixed-width decimal/binary coding and its partial decoder, language restriction, individual-word refusal, erasure, renaming inverses and arrow/trace reversal. The discrete-category companion constructs functors and natural isomorphisms, deriving an object bijection from an equivalence and the finite obstruction for two versus twenty-six objects.

The Python tests check report reconstruction, disjoint demand accounting, tagging, graph ranks, integral quotient classes, code collisions, CHSH signs and source mutations. Rust tests check the corresponding runtime examples and malformed inputs, free-word cancellation and exact joint mixtures. Fresh Lean, Python and Rust executions agree on 33,104 versioned replay records, including 13,493 refusals. The reproduction uses canonical ASCII decimal spellings, an eight-element carrier/history bound, at most sixteen graph edges and integer magnitude at most one million, signed words of length at most 64 and joint mixtures with sixteen nonnegative integer masses. It is a finite comparison, not a general compiler-refinement theorem. Grammar conformance covers 30 Pre-Genesis, 85 S(n) and 39 MSS-N recorded controls, without supplying semantic evaluators for every grammar start.

This remains an author review draft. Local compilation, executable checks and package integrity do not discharge independent mathematical review or authorize a public deposit. No new model trial, physical validation or general derivation of the MSS atlas is claimed.

Appendix A. The 22 declared MSS grammar starts

This table is the source inventory, not a list of completed semantic adapters. The production grammar remains the authority for accepted syntax.

#	Start symbol	Canonical input	Lexical and structural contract
01	raw_glyphs	`a><b0<<`	Raw strings over Γ_{38}, including ε. No balancing required.
02	word	alpha01	Nonempty sequence over Σ_{36}. No delimiters.
03	glyph_path	`a>b<c>a`	Alternating single glyphs and directions.
04	label_path	`a0>b2>c1`	Directed paths between multi-glyph identifiers.
05	suffix_path	`a0>b12<c3`	Vertex structured as lower_latin followed by digits.
06	path_list	`a>b;c<d`	Semicolon-delimited sequence of independent paths.
07	suffix_list	`a0>b1;c2<d3`	Semicolon-delimited sequence of suffix paths.
08	scalar_path	`a:1.5>b:-2/3`	Paths with colon-separated scalar weights. `a:1/0` parses syntactically.
09	vector_path	`a[1,2]>b[3,4]`	Bracketed vector-annotated vertices.
10	edge_weight_path	`a>2.5>b<1<c`	Arrow-flanked edge weights (`>w>` or `<w<`). Opposing marks (`>w<`) rejected.
11	labelled_path	`a-step>b`	Named relation links (`-label>`).
12	prefix_tree	`>ab`	Polish prefix notation for binary syntax trees.
13	postfix_tree	`ab>`	Reverse Polish notation for binary syntax trees.
14	sequence	`a,b,c`	Comma-delimited list of single glyphs.
15	nested_tree	`(a,(b,c))`	Parenthesized nested trees.
16	head_term	`f(x,g(y))`	Functional terms with head words and argument lists.
17	incidence	`[a,b]>[c,d]`	Hypergraph directed incidence lists.
18	unicode_path	$\alpha\rightarrow\beta\leftarrow\gamma$	Directed paths over Greek/symbolic Unicode alphabets.
19	arrow_path	a→b←c	ASCII glyphs joined by Unicode arrows (→, ←).
20	bitstream	10011010	Nonempty binary digit strings (0 and 1).
21	octets	deadbeef	Lowercase hexadecimal byte strings (even length required).
22	rewrite_rule	`a=>b`	String rewrite rules with optional empty sides.